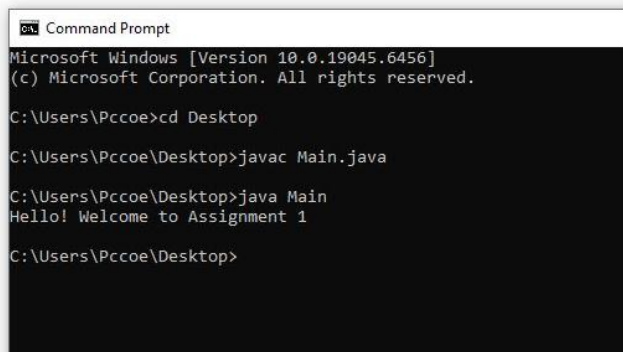


PRACTICE PROGRAMS:

1. VARIABLES, DATA TYPES, OPERATORS, AND INPUT/OUTPUT

Program 1- Hello world

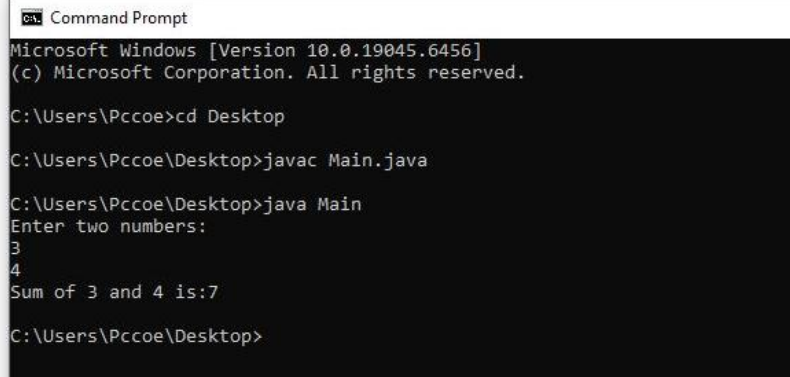
```
public class Main{  
    public static void main(String args[]){  
        System.out.println("Hello! Welcome to Assignment 1");  
    }  
}
```



```
Command Prompt  
Microsoft Windows [Version 10.0.19045.6456]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Pccoe>cd Desktop  
  
C:\Users\Pccoe\Desktop>javac Main.java  
  
C:\Users\Pccoe\Desktop>java Main  
Hello! Welcome to Assignment 1  
  
C:\Users\Pccoe\Desktop>
```

Program 2- Addition of two numbers

```
import java.util.*;  
public class Main{  
    public static void main(String args[]){  
        Scanner sc=new Scanner(System.in);  
        int a,b,sum=0;  
        System.out.println("Enter two numbers:");  
        a=sc.nextInt();  
        b=sc.nextInt();  
        sum=a+b;  
        System.out.println("Sum of "+a+" and "+b+" is:"+sum);  
    }  
}
```

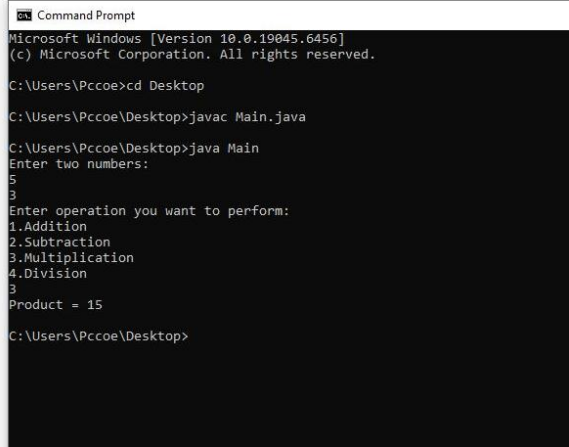


```
Command Prompt  
Microsoft Windows [Version 10.0.19045.6456]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Pccoe>cd Desktop  
  
C:\Users\Pccoe\Desktop>javac Main.java  
  
C:\Users\Pccoe\Desktop>java Main  
Enter two numbers:  
3  
4  
Sum of 3 and 4 is:7  
  
C:\Users\Pccoe\Desktop>
```

2. CONDITIONAL STATEMENTS

Program 3- Calculator program

```
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int a,b,choice=0;
        System.out.println("Enter two numbers:");
        a=sc.nextInt();
        b=sc.nextInt();
        System.out.println("Enter operation you want to perform:\n1.Addition\n2.Subtraction\n3.Multiplication\n4.Division");
        choice=sc.nextInt();
        switch(choice){
            case 1:
                System.out.println("Sum = "+(a+b));
                break;
            case 2:
                System.out.println("Difference = "+(a-b));
                break;
            case 3:
                System.out.println("Product = "+(a*b));
                break;
            case 4:
                System.out.println("Quotient = "+(a/b));
                break;
            default:
                System.out.println("Invalid Choice.");
        }
    }
}
```

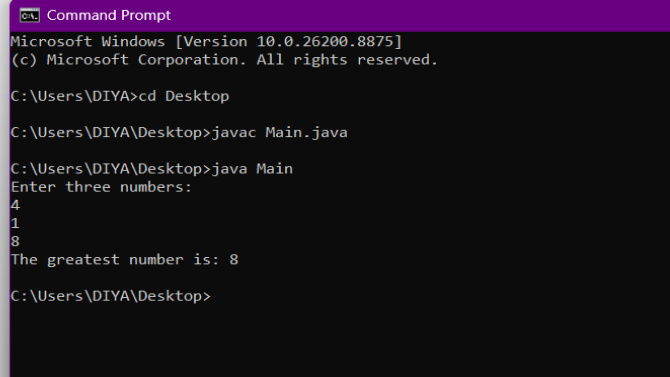


```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Pccoe>cd Desktop
C:\Users\Pccoe\Desktop>javac Main.java
C:\Users\Pccoe\Desktop>java Main
Enter two numbers:
5
3
Enter operation you want to perform:
1.Addition
2.Subtraction
3.Multiplication
4.Division
3
Product = 15
C:\Users\Pccoe\Desktop>
```

Program 4- Greatest of three integers

```
import java.util.*;
class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int a,b,c;
        System.out.println("Enter three numbers:");
        a=sc.nextInt();
        b=sc.nextInt();
        c=sc.nextInt();
        if (a>=b && a>=c) {
            System.out.println("The greatest number is: " +a);
        } else if (b>=a && b>=c) {
            System.out.println("The greatest number is: " +b);
        } else {
            System.out.println("The greatest number is: " +c);
        }
    }
}
```



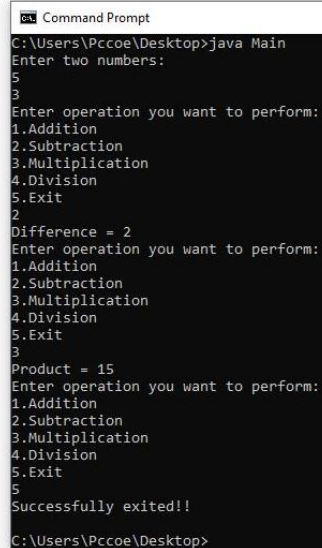
```
Command Prompt
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DIYA>cd Desktop
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter three numbers:
4
1
8
The greatest number is: 8
C:\Users\DIYA\Desktop>
```

3. LOOPS (FOR, WHILE, AND DO-WHILE)

Program 5- Calculator with do-while loop

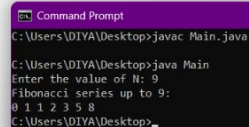
```
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int a,b,choice=0;
        System.out.println("Enter two numbers:");
        a=sc.nextInt();
        b=sc.nextInt();
        do{
            System.out.println("Enter operation you want to perform:\n1.Addition\n2.Subtraction\n3.Multiplication\n4.Division\n5.Exit");
            choice=sc.nextInt();
            switch(choice){
                case 1:
                    System.out.println("Sum = "+(a+b));
                    break;
                case 2:
                    System.out.println("Difference = "+(a-b));
                    break;
                case 3:
                    System.out.println("Product = "+(a*b));
                    break;
                case 4:
                    System.out.println("Quotient = "+(a/b));
                    break;
                case 5:
                    System.out.println("Successfully exited!!");
                    break;
                default:
                    System.out.println("Invalid Choice.");
                    break;
            }
        } while(choice!=5);
    }
}
```



```
Command Prompt
C:\Users\Pccoe\Desktop>java Main
Enter two numbers:
5
3
Enter operation you want to perform:
1.Addition
2.Subtraction
3.Multiplication
4.Division
5.Exit
2
Difference = 2
Enter operation you want to perform:
1.Addition
2.Subtraction
3.Multiplication
4.Division
5.Exit
3
Product = 15
Enter operation you want to perform:
1.Addition
2.Subtraction
3.Multiplication
4.Division
5.Exit
5
Successfully exited!!
C:\Users\Pccoe\Desktop>
```

Program 6- Fibonacci Series

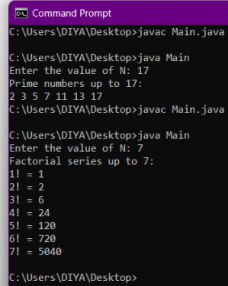
```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the value of N: ");
        int n = sc.nextInt();
        System.out.println("Fibonacci series up to " + n + ":");
        int a=0;
        int b=1;
        while(a<=n) {
            System.out.print(a+" ");
            b=a+b;
            a=b-a;
        }
    }
}
```



```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter the value of N: 9
Fibonacci series up to 9:
0 1 1 2 3 5 8
C:\Users\DIYA\Desktop>
```

Program 7- Factorial Series

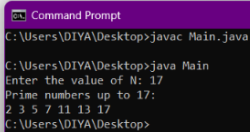
```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the value of N: ");
        int n = sc.nextInt();
        System.out.println("Factorial series up to " + n + ":");
        int fact= 1;
        for (int i = 1; i <= n; i++) {
            fact*= i;
            System.out.println(i + "! = " + fact);
        }
    }
}
```



```
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter the value of N: 17
Prime numbers up to 17:
2 3 5 7 11 13 17
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter the value of N: 7
Factorial series up to 7:
1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
6! = 720
7! = 5040
C:\Users\DIYA\Desktop>
```

Program 8- Prime number series

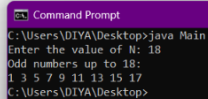
```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the value of N: ");
        int n = sc.nextInt();
        System.out.println("Prime numbers up to " + n + ":");
        for (int i = 2; i <= n; i++) {
            int check=0;
            for (int j = 2; j * j <= i; j++) {
                if (i % j == 0) {
                    check=1;
                    break;
                }
            }
            if (check==0) {
                System.out.print(i + " ");
            }
        }
    }
}
```



```
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter the value of N: 17
Prime numbers up to 17:
2 3 5 7 11 13 17
C:\Users\DIYA\Desktop>
```

Program 9- Odd number series

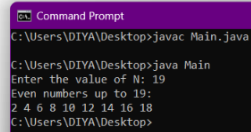
```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the value of N: ");
        int n = sc.nextInt();
        System.out.println("Odd numbers up to " + n + ":");
        for (int i = 1; i <= n; i += 2) {
            System.out.print(i + " ");
        }
    }
}
```



```
C:\Users\DIYA\Desktop>java Main
Enter the value of N: 18
Odd numbers up to 18:
1 3 5 7 9 11 13 15 17
C:\Users\DIYA\Desktop>
```

Program 10- Even number series

```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the value of N: ");
        int n = sc.nextInt();
        System.out.println("Even numbers up to " + n + ":");
        for (int i = 2; i <= n; i += 2) {
            System.out.print(i + " ");
        }
    }
}
```

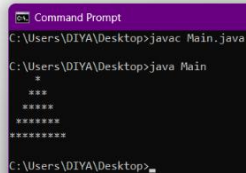


```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter the value of N: 19
Even numbers up to 19:
2 4 6 8 10 12 14 16 18
C:\Users\DIYA\Desktop>
```

4. PATTERNS

Program 11-

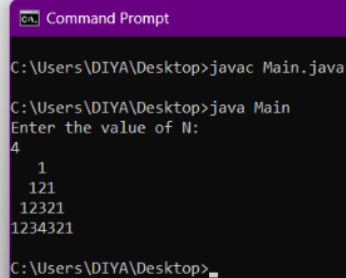
```
import java.util.*;
class Main{
    public static void main(String[] args) {
        int rows = 5;
        for (int i = 1; i <= rows; i++) {
            for (int j = i; j <= rows; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= (2 * i - 1); k++) {
                System.out.print("**");
            }
            System.out.println();
        }
    }
}
```



```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
      *
     **
    ***
   ****
  *****
 *****
  *****
   ****
    ***
     **
      *
```

Program 12-

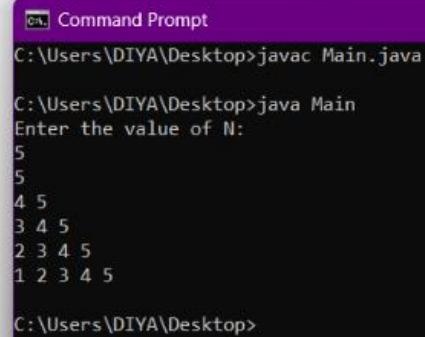
```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the value of N:");
        int n=sc.nextInt();
        for(int i=0;i<n;i++){
            for(int j=0;j<n-i-1;j++){
                System.out.print(" ");
            }
            int k=1;
            int breakpoint=(2*i+1)/2;
            for(int j=1;j<=2*i+1;j++){
                System.out.print(k);
                if(j<=breakpoint) k++;
                else k--;
            }
            for(int j=0;j<n-i-1;j++){
                System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```



```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter the value of N:
4
      1
     121
    12321
   1234321
C:\Users\DIYA\Desktop>
```

Program 13-

```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the value of N:");
        int n=sc.nextInt();
        for(int i=0;i<n;i++){
            for(int j=n-i;j<=n;j++){
                System.out.print(j+" ");
            }
            System.out.println();
        }
    }
}
```



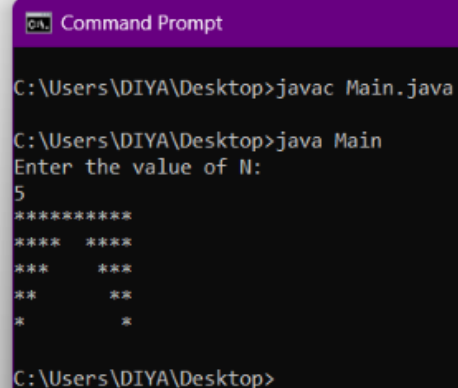
```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java

C:\Users\DIYA\Desktop>java Main
Enter the value of N:
5
5
4 5
3 4 5
2 3 4 5
1 2 3 4 5

C:\Users\DIYA\Desktop>
```

Program 14-

```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the value of N:");
        int n=sc.nextInt();
        int k=0;
        for(int i=0;i<n;i++){
            for(int j=1;j<=n-i;j++){
                System.out.print("*");
            }
            for(int j=0;j<k;j++){
                System.out.print(" ");
            }
            for(int j=1;j<=n-i;j++){
                System.out.print("*");
            }
            k+=2;
            System.out.println();
        }
    }
}
```



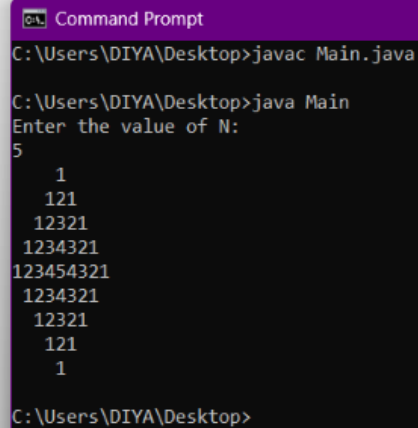
```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java

C:\Users\DIYA\Desktop>java Main
Enter the value of N:
5
*****
***  ***
**   **
*    *
*****

C:\Users\DIYA\Desktop>
```

Program 15-

```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the value of N:");
        int n=sc.nextInt();
        for(int i=0;i<n;i++){
            for(int j=1;j<=n-i-1;j++){
                System.out.print(" ");
            }
            int k=1;
            int b=(2*i+1)/2;
            for(int j=1;j<=2*i+1;j++){
                System.out.print(k);
                if(j<=b)k++;
                else k--;
            }
            for(int j=0;j<n-i-1;j++){
                System.out.print(" ");
            }
            System.out.println();
        }
        for(int i=1;i<n;i++){
            for(int j=1;j<=i;j++){
                System.out.print(" ");
            }
            int k=1;
            int b=(2*(n-i)-1)/2;
            for(int j=1;j<=2*(n-i)-1;j++){
                System.out.print(k);
                if(j<=b)k++;
                else k--;
            }
            System.out.println();
        }
    }
}
```

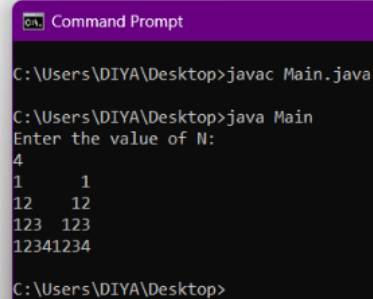


```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter the value of N:
5
    1
   121
  12321
 1234321
123454321
 1234321
  12321
   121
    1

C:\Users\DIYA\Desktop>
```

Program 16-

```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the value of N:");
        int n=sc.nextInt();
        int t=2*(n-1);
        for(int i=1;i<=n;i++){
            for(int j=1;j<=i;j++){
                System.out.print(j);
            }
            for(int j=0;j<t;j++){
                System.out.print(" ");
            }
            for(int j=1;j<=i;j++){
                System.out.print(j);
            }
            System.out.println();
            t-=2;
        }
    }
}
```



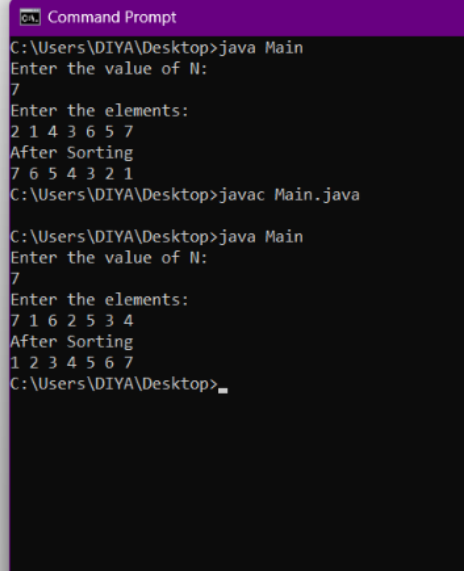
```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter the value of N:
4
    1    1
   12   12
  123  123
 12341234

C:\Users\DIYA\Desktop>
```

5. ARRAYS

Program 17-Bubble Sort

```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the value of N:");
        int n=sc.nextInt();
        int[] arr=new int[n];
        System.out.println("Enter the elements:");
        for(int i=0;i<n;i++){
            arr[i]=sc.nextInt();
        }
        for(int i=n-1;i>=0;i--){
            for(int j=0;j<i;j++){
                if(arr[i]<arr[j]){
                    int t=arr[i];
                    arr[i]=arr[j];
                    arr[j]=t;
                }
            }
        }
        System.out.println("After Sorting");
        for(int i=0;i<n;i++){
            System.out.print(arr[i]+" ");
        }
    }
}
```

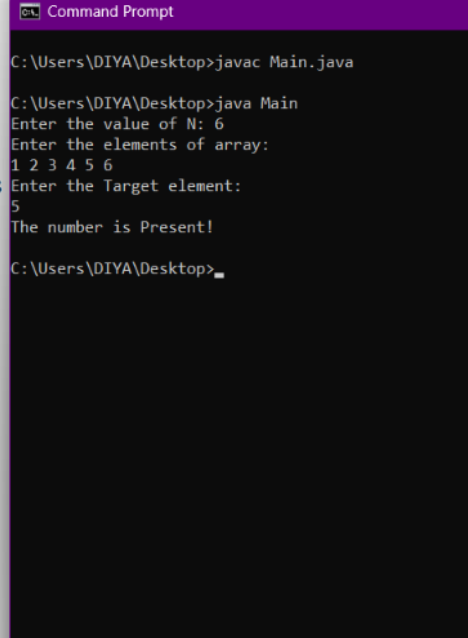


```
Command Prompt
C:\Users\DIYA\Desktop>java Main
Enter the value of N:
7
Enter the elements:
2 1 4 3 6 5 7
After Sorting
7 6 5 4 3 2 1
C:\Users\DIYA\Desktop>javac Main.java

C:\Users\DIYA\Desktop>java Main
Enter the value of N:
7
Enter the elements:
7 1 6 2 5 3 4
After Sorting
1 2 3 4 5 6 7
C:\Users\DIYA\Desktop>
```

Program 18- Linear Search

```
import java.util.*;
class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the value of N: ");
        int n = sc.nextInt();
        int [] arr=new int[n];
        System.out.println("Enter the elements of array:");
        for(int i=0;i<n;i++){
            arr[i]=sc.nextInt();
        }
        System.out.println("Enter the Target element:");
        int t=sc.nextInt();
        int p=0;
        for(int i=0;i<n;i++){
            if(arr[i]==t)
                p=1;
        }
        if(p==1)
            System.out.println("The number is Present!");
        else
            System.out.println("The number is not present!");
    }
}
```



```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java

C:\Users\DIYA\Desktop>java Main
Enter the value of N: 6
Enter the elements of array:
1 2 3 4 5 6
Enter the Target element:
5
The number is Present!
C:\Users\DIYA\Desktop>
```

6. STRINGS

Program 19- Palindrome String

```
import java.util.*;
public class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

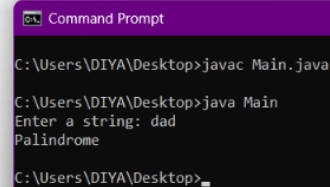
        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String rev = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            rev += str.charAt(i);
        }

        if (str.equalsIgnoreCase(rev))
            System.out.println("Palindrome");
        else
            System.out.println("Not a Palindrome");

        sc.close();
    }
}
```



```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter a string: dad
Palindrome
C:\Users\DIYA\Desktop>
```

Program 20- Number of Vowels, Digits, Consonants and Spaces String

```
import java.util.*;
public class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

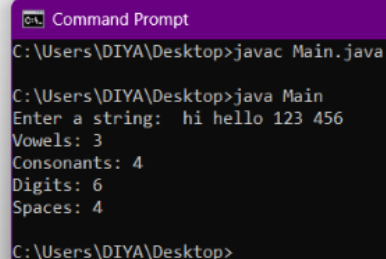
        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        int vowels = 0, consonants = 0, digits = 0, spaces = 0;

        for (int i = 0; i < str.length(); i++) {
            char ch = Character.toLowerCase(str.charAt(i));

            if (ch >= 'a' && ch <= 'z') {
                if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                    vowels++;
                else
                    consonants++;
            } else if (ch >= '0' && ch <= '9') {
                digits++;
            } else if (ch == ' ') {
                spaces++;
            }
        }

        System.out.println("Vowels: " + vowels);
        System.out.println("Consonants: " + consonants);
        System.out.println("Digits: " + digits);
        System.out.println("Spaces: " + spaces);
    }
}
```



```
Command Prompt
C:\Users\DIYA\Desktop>javac Main.java
C:\Users\DIYA\Desktop>java Main
Enter a string: hi hello 123 456
Vowels: 3
Consonants: 4
Digits: 6
Spaces: 4
C:\Users\DIYA\Desktop>
```